

Executive Summary

Patient Monitoring via Structured Electronic Health Records

Client: SwissCare Health Network AG

Timeline: 22 weeks

SwissCare Health Network manages numerous hospitals and affiliated clinics across Switzerland and the EU, processing over 3 million patient encounters annually. Current EHR systems contain 70% unstructured clinical notes that remain inaccessible for systematic analysis, leading to missed opportunities for early intervention and care coordination failures. Digital health leaders are capturing efficiency gains through AI-powered patient monitoring systems. Meanwhile, evolving regulations under the EU AI Act, GDPR/revFADP, and Swiss healthcare data protection laws require enhanced transparency, continuous monitoring, and robust governance for AI-driven clinical decision support systems.

We have been tasked with developing an end-to-end machine learning pipeline that transforms unstructured electronic health records into standardized patient profiles with automated risk monitoring capabilities. Deliverables include a data transformation system for tabular data, clinical text, an AI monitoring framework for predictive risk assessment, and documentation with necessary compliance information for high-risk medical AI applications.

Business Challenge

- **Data Fragmentation:** A large percentage of clinical information is documented only in unstructured notes, making it virtually inaccessible for care decisions
- **Care Coordination Failures:** hospital readmissions result from information gaps between providers; there are not enough resources to provide the manual reviews currently needed to prevent such situations
- **Regulatory Exposure:** Non-compliance with medical data regulations carries huge penalties

Objectives

1. **Automate Data Structuring:** Enable automatic conversion of unstructured clinical notes to entries in standardized, structured profiles
2. **Enable Proactive Monitoring:** Identify high-risk patients at least 48-72 hours before adverse events without the need for extensive manual review
3. **Ensure Compliance:** Full adherence to EU AI Act, Swiss Federal Act on Data Protection (revFADP), and healthcare standards

Data Assets:

- EHR data in CSV format with 50,000+ patient records

- Clinical notes in multiple formats (free text, semi-structured reports)

Regulatory & Compliance Requirements

EU AI Act (High-Risk Medical AI):

AI is **high-risk** when it is a component (including a safety component) of a medical device; high-risk AI must have documented risk management, suitable data governance, clear human oversight, technical documentation, accuracy/robustness/cybersecurity, logging, and post-market monitoring.

- **Model Development & Testing:** You must keep documentation describing the intended purpose, limits, known risks, performance metrics, and how the system is meant to be used. You must test for accuracy and robustness before release and keep evidence
- **Deployment & Operations:** Systems must be designed so people can intervene and prevent or minimize harm. Event logging and post-market monitoring are required to track real-world performance and incidents

GDPR/revFADP (Health Data Protection):

When a tool solely makes decisions that have legal or similarly significant effects on a person (e.g., care eligibility), people have protections: the right not to be subject to such a decision, and—where such decisions are allowed—meaningful information, human review, and a way to contest; Switzerland's revFADP mirrors this with a duty to inform and enable human review of automated decisions.

- **Model Development & Testing:** Where automated decisions are used, people must receive meaningful information about the logic involved and the likely consequences

HL7 FHIR Compliance:

The eHealth Network (the official EU coordination body for eHealth) recommends HL7 FHIR as the base standard for cross-border health data exchange and publishes implementation guidance; aligning with FHIR resources and schemas helps systems interoperate across EU programs.

- **Data Interoperability:** Healthcare data exchange requires particular resource formatting (called FHIR; we will use a simplified version of this form). All data must be stored in this format in order to be used across systems

SNOMED CT/ICD-10 Standards:

ICD-10 is the international classification for diseases/events; SNOMED CT is a clinical terminology used to capture clinical concepts with precision; health programs often require them and many countries license SNOMED via national agreements

- **Clinical Coding Accuracy:** Records about patients’ diagnoses and procedures require the use of this standard terminology mapping to be compatible with other healthcare organizations/systems.

SwissCare Health Network AG and Hospital Policies

- **Clinical Safety Policy:** SwissCare Health Network AG has promised hospitals that alerts from this tool will be “actionable and understandable to clinicians making urgent decisions.”
- **Clinical Accountability Policy:** Several of SwissCare Health Network AG’s clients have made clear that its “clinicians must know the rationale behind patient-risk alerts before acting.”

Stakeholders and Their Priorities

Stakeholder	Primary Concerns	Success Metrics
Chief Medical Officer	Patient safety, clinical accuracy, interpretability	Adverse event prediction rate, false positive rate
Chief Information Officer	System reliability, interoperability	Uptime, formatting compliance rate
Compliance Officer	Regulatory adherence, audit readiness	Zero violations, documentation completeness
Emergency Department Director	Early warning accuracy, actionable alerts	Alert recall and precision
Nursing Leadership	Usability, alert fatigue	Interpretability of alerts, false positive alert rate
Patients	Care quality, accuracy	Outcome improvements

Project Implementation

Solution Components

Through this project, you will gain experience in some of the methods needed for building a successful AI-backed healthcare system. Concretely, you will implement the following pipeline components:

Clinical Text Processing Pipeline

- Transform unstructured clinical notes from EHRs into structured patient data, handling medical terminology, abbreviations, and temporal references
- Extract and normalize temporal information to construct accurate patient timelines from narrative clinical notes
- Map extracted entities to standard medical vocabularies (ICD-10, SNOMED) for interoperability

Patient Profile Construction System

- Build standardized patient profiles following FHIR-based structures that consolidate all patient information
- Create chronological event timelines integrating data from multiple sources (structured records and extracted text)
- Engineer clinically meaningful features including vital sign trends, comorbidity indicators, and risk factor combinations
- Implement temporal feature engineering that prevents data leakage while capturing patient history

Risk Prediction Framework

- Develop binary classifiers to predict 30-day readmission or adverse events using historical patient data, using multiple modeling approaches from traditional ML (XGBoost) to temporal models (LSTMs) for sequence analysis
- Create training datasets with proper patient-level splitting and temporal validation strategies

Monitoring Workflow System

- Design autonomous monitoring workflow that continuously assesses patient risk scores as new data arrives
- Implement automated alert generation for high-risk patients with configurable thresholds
- Build simulation environments to test system behavior under various clinical scenarios

Project-Specific Grading Details

Technical Requirements

Pipeline Assessment Criteria (What instructors will verify):

The final pipeline must demonstrate the following capabilities when tested with new patient data:

- **End-to-end processing:** Accept raw EHR data (CSV and text) and output structured patient profiles
- **NLP accuracy:** Successfully extract and categorize medical entities from clinical notes, populating patient profiles with this information
- **Timeline construction:** Generate chronologically accurate patient event sequences from mixed data sources
- **Risk prediction:** Functional models predicting readmission with AUROC > baseline threshold
- **Automated checks:** Automated monitoring system that triggers alerts based on risk thresholds

Example Advanced Capabilities (For extension credit):

- Advanced text processing techniques for extracting structured information from unstructured notes data
- Ensemble methods for risk prediction
- Interpretability framework providing natural language explanations for risk classifications

Report Requirements

Aside from the general requirements described in the Grading Rubric, for this project, it is expected that your report contain the following information:

Clinical Validity Assessment

- **Risk Prediction Calibration:** Provide alert performance analysis, e.g., for every 100 alerts, how many readmissions were prevented vs. false alarms?
- **Demographic Fairness:** Model performance across age, gender, and ethnicity groups

Regulatory Compliance Assessment

- **GDPR/revFADP Article 22 (Automated Decisions):** Document the human review process - when does a clinician intervene and what information are they provided?
- **EU AI Act High-Risk Requirements:** Explain your data quality measures and accuracy testing approach for this medical application